Assumptions Leading to No Patterns in Hopalong Attractors

1. Zero Parameters

When all parameters a, b, and c are zero, the equations simplify significantly, leading to trivial or no patterns.

$$x_{n+1} = y_n - \text{sgn}(x_n)\sqrt{|0 \cdot x_n - 0|} = y_n$$
$$y_{n+1} = 0 - x_n = -x_n$$

In this case, the points oscillate between (x, y) and $(y, -x)$ without generating complex patterns.

2. Parameters Leading to Fixed Points

Certain parameter combinations cause the system to converge to a fixed point or a simple limit cycle, preventing the formation of complex attractors.

For example, when $a = 0$ and $b = 0$:

$$x_{n+1} = y_n - \text{sgn}(x_n)\sqrt{|0 \cdot x_n - c|} = y_n - \text{sgn}(x_n)\sqrt{|c|}$$
$$y_{n+1} = 0 - x_n = -x_n$$

Depending on the value of c, this can lead to a fixed point or a simple oscillation, not an attractor.

3. Symmetrical Simplifications

Parameters that lead to symmetrical or non-chaotic behavior result in simple or repetitive patterns.

For instance, when $a = 1$, $b = 0$, and $c = 1$:

$$x_{n+1} = y_n - \text{sgn}(x_n)\sqrt{|0 \cdot x_n - 1|} = y_n - \text{sgn}(x_n)$$
$$y_{n+1} = 1 - x_n$$

Here, the dynamics might settle into non-complex patterns or fixed points due to the lack of chaotic interactions.

4. Degenerate Dynamics

Certain parameter values reduce the equations to simple forms that do not support chaotic behavior.

For example, when $a = 0$, $b = 1$, and $c = 0$:

$$x_{n+1} = y_n - \text{sgn}(x_n)\sqrt{|1 \cdot x_n - 0|} = y_n - \text{sgn}(x_n)\sqrt{|x_n|}$$
$$y_{n+1} = 0 - x_n = -x_n$$

This scenario leads to simple harmonic motion, tracing basic shapes like lines or circles.

5. Simplifications with $b = 0$

When $b = 0$, the attractor equations simplify, and the resulting behavior is often non-chaotic, even if a and c have different values.

For instance, with $b = 0$ and any a and c:

$$x_{n+1} = y_n - \text{sgn}(x_n)\sqrt{|0 \cdot x_n - c|} = y_n - \text{sgn}(x_n)\sqrt{|c|}$$
$$y_{n+1} = a - x_n$$

Since the term involving x_n in the square root is absent, the equations do not exhibit the sensitivity and complexity typical of chaotic systems, leading to simple and repetitive patterns.

Why These Parameters Fail to Produce Patterns

1. **Lack of Chaos:** Chaotic attractors require sensitive dependence on initial conditions, leading to complex and seemingly random patterns. Parameters leading to simple harmonic motion or fixed points lack the necessary chaotic behavior.

2. **Rapid Convergence:** If the system quickly converges to a stable point or a simple cycle, it does not explore a wide range of the phase space, resulting in a lack of complex structure.

3. **Overly Simplified Dynamics:** Parameters that reduce the system to linear or otherwise simple equations do not allow for the intricate interactions needed to produce chaotic attractors.

Detailed Analysis of Parameters $a = 1$, $b = 2$, and $c = 3$

The Hopalong attractor equations with these parameters are:

$$x_{n+1} = y_n - \text{sgn}(x_n)\sqrt{|2x_n - 3|}$$
$$y_{n+1} = 1 - x_n$$

Initial Conditions

Typically, initial conditions are set to small values (e.g., $x_0 = 0$ and $y_0 = 0$).

Iteration Behavior

- For x_n close to zero, the term $|2x_n - 3|$ is dominated by $|3|$.
- As x_n increases, $|2x_n - 3|$ grows, causing rapid changes in x_n and y_n.

Possible Convergence to Non-Patterned Behavior

- The equations might lead to x_n and y_n values that oscillate or converge in such a way that the attractor does not explore a wide range of the plane.
- Specifically, if x_n and y_n fall into a repetitive cycle or a very tight range, the attractor will not exhibit typical chaotic behavior.

Conclusion

By avoiding degenerate or overly simplistic parameter choices, one can typically ensure that the Hopalong attractor generates the complex and interesting patterns it is known for.

Reference: ChatGPT

```
In[53]:= a = 1;
b = 2;
c = 3;

nMax = 100; (*Number of iterations*)

(*Define the compiled function*)
cf = Compile[{{x, _Real}, {y, _Real}}, {y -
    Sign[x]  Sqrt[Abs[b  x - c]], a - x}, CompilationTarget -> "C",
  RuntimeOptions -> "Speed"];

(*Initialize x and y*)
x = 0.;
y = 0.;

(*Preallocate lists*)
xValues = Table[0., {nMax + 1}];
yValues = Table[0., {nMax + 1}];

(*Initial values*)
xValues[[1]] = x;
yValues[[1]] = y;

(*Perform the iterations*)
Do[{x, y} = cf[x, y];
  xValues[[n + 1]] = x;
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  yValues[[n + 1]] = y, {n, 1, nMax}];

(*Output results*)
{xValues, yValues}

Out[65]= {{0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1.,
   0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0.,
   1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0.,
   0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1.,
   0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0.,
   1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0., 0., 1., 0.,
   0.}, {0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1.,
  0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1.,
  1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0.,
  1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1.,
  0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1.,
  1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0.,
  1.}}

In[66]:= a = 1;
```

```
b = 2;
c = 3.000001;

nMax = 100; (*Number of iterations*)

(*Define the compiled function*)
cf = Compile[{{x, _Real}, {y, _Real}}, {y -
    Sign[x]  Sqrt[Abs[b  x - c]], a - x}, CompilationTarget -> "C",
  RuntimeOptions -> "Speed"];

(*Initialize x and y*)
x = 0.;
y = 0.;

(*Preallocate lists*)
xValues = Table[0., {nMax + 1}];
yValues = Table[0., {nMax + 1}];

(*Initial values*)
xValues[[1]] = x;
yValues[[1]] = y;

(*Perform the iterations*)
Do[{x, y} = cf[x, y];
  xValues[[n + 1]] = x;
  yValues[[n + 1]] = y, {n, 1, nMax}];
```

```
(*Output results*)
{xValues, yValues}

Out[78]= {{0., 0., 1., -5.*10^-7, 1.73205, 0.31875, -2.26909, 3.42683,
   1.30602, -3.04969, 2.71049, 2.49373, -3.12027, 1.54609,
  3.81666, -2.6986, 0.0811375, 2.01405, -0.0950888,
  0.772058, -0.111512, 2.02322, 0.0885553, -2.70337, 3.81088,
  1.55354, -3.1381, 2.49215, 2.72945, -3.06024, 1.29056,
  3.41303, -2.24659, 0.324339, 1.71319, 0.0226872, -2.43209, 3.78163,
  1.29591, -3.42051, 2.84113, 2.78275, -3.44285, 1.3614,
  3.91635, -2.55974, -0.0668779, 5.32998, -1.70079, -1.79985, 5.26977,
   0.0540242, -5.97035, 4.8113, 4.39691, -6.21833, 0.532044,
  5.82696, -2.4738, -2.00781, 6.1225, -0.0327487, -3.37164, 4.15417,
  2.06766, -4.21969, 2.31455, 3.94332, -3.52513, 0.226894,
  2.92944, -0.917719, 0.269519, 0.348973, -0.786772, 2.78961,
  0.180776, -3.41394, 3.95417, 2.19846, -4.13609, 2.15894,
  3.9881, -3.38968, 0.1391, 2.73989, -0.713832, 0.364311,
  0.206722, -0.972588, 3.01705, 0.23072, -3.61034, 3.96626,
  2.38941, -4.29999, 2.01646, 4.28366, -3.37598, -0.160846,
  6.19853}, {0., 1., 1., 0., 1., -0.732051, 0.68125,
  3.26909, -2.42683, -0.306022, 4.04969, -1.71049, -1.49373,
  4.12027, -0.546091, -2.81666, 3.6986, 0.918862, -1.01405, 1.09509,
  0.227942, 1.11151, -1.02322, 0.911445, 3.70337, -2.81088, -0.553538,
   4.1381, -1.49215, -1.72945, 4.06024, -0.290559, -2.41303, 3.24659,
  0.675661, -0.713188, 0.977313, 3.43209, -2.78163, -0.295913,
```

4.42051, -1.84113, -1.78275, 4.44285, -0.361399, -2.91635, 3.55974, 1.06688, -4.32998, 2.70079, 2.79985, -4.26977, 0.945976, 6.97035, -3.8113, -3.39691, 7.21833, 0.467956, -4.82696, 3.4738, 3.00781, -5.1225, 1.03275, 4.37164, -3.15417, -1.06766, 5.21969, -1.31455, -2.94332, 4.52513, 0.773106, -1.92944, 1.91772, 0.730481, 0.651027, 1.78677, -1.78961, 0.819224, 4.41394, -2.95417, -1.19846, 5.13609, -1.15894, -2.9881, 4.38968, 0.8609, -1.73989, 1.71383, 0.635689, 0.793278, 1.97259, -2.01705, 0.76928, 4.61034, -2.96626, -1.38941, 5.29999, -1.01646, -3.28366, 4.37598, 1.16085}}